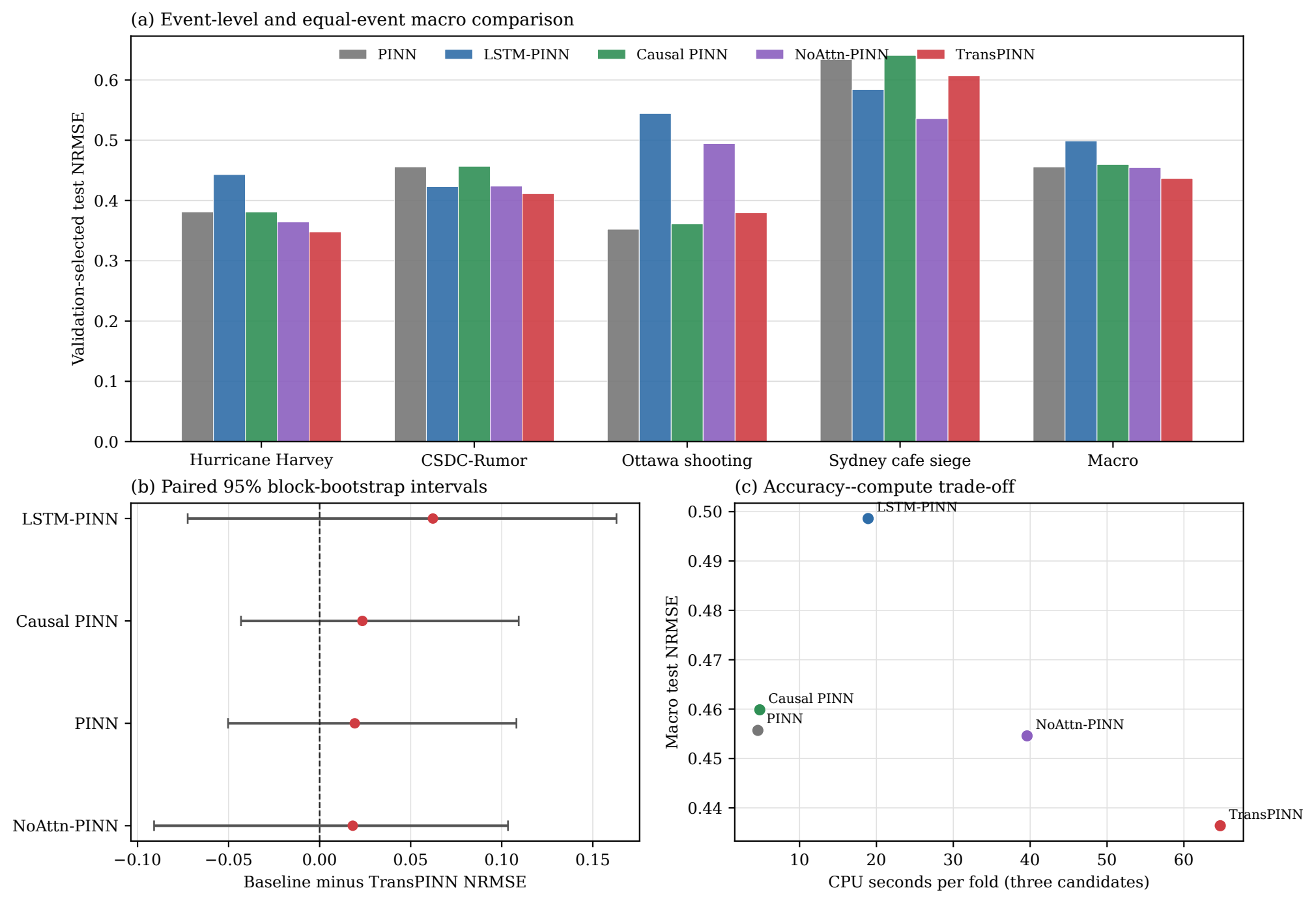
(a) Event-level and equal-event macro comparison
PINN
LSTM-PINN
Causal PINN
NoAttn-PINN
TransPINN
Validation-selected test NRMSE
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Hurricane Harvey
CSDC-Rumor
Ottawa shooting
Sydney cafe siege
Macro
(b) Paired 95% block-bootstrap intervals
LSTM-PINN
Causal PINN
PINN
NoAttn-PINN
−0.10
−0.05
0.00
0.05
0.10
0.15
Baseline minus TransPINN NRMSE
(c) Accuracy--compute trade-off
Macro test NRMSE
0.44
0.45
0.46
0.47
0.48
0.49
0.50
LSTM-PINN
Causal PINN
PINN
NoAttn-PINN
TransPINN
10
20
30
40
50
60
CPU seconds per fold (three candidates)